

Occupancy Sensor API Specification

Required Endpoints

GET /{building}/{floor}/sensors

Returns all sensor names on a given floor in a given building.

Response — 200 OK

```
{
  "building": "Al-Midra Tower",
  "floor": 3,
  "sensors": [
    "Sensor1",
    "Sensor2",
    "Sensor3"
  ]
}
```

Error responses

HTTP status	error value	Meaning
404	"floor_not_found"	No floor with that number exists in the specified building
404	"building_not_found"	No building with that name exists

```
{ "error": "floor_not_found", "message": "Floor 3 not found in Al-Midra Tower." }
```

GET /{building}/{floor}/{sensor_name}

Returns the layout and occupancy data for a specific sensor.

Response — 200 OK

```
{
  "sensor_name": "Sensor1",
  "bounding_box": {
    "x1": 400,
    "y1": 180,
    "x2": 520,
    "y2": 300
  },
  "seats": [
    {
      "seat_id": "Sensor1-001",
      "x": 412,
      "y": 208,
      "assigned": true,
      "organization_code": "FIN",
      "organization_name": "Finance"
    },
    {
      "seat_id": "Sensor1-002",
      "x": 445,
      "y": 208,
      "assigned": false,
      "organization_code": null,
      "organization_name": null
    }
  ]
}
```

Response fields

Top-level

Field	Type	Description
sensor_name	string	Echo of the requested sensor name
bounding_box	object	Pixel coordinates of the top-left (x1, y1) and bottom-right (x2, y2) corners of the area covered by this sensor

<code>seats</code>	array	All seats within this sensor's coverage area.
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`bounding_box` object

Field	Type	Description
<code>x1</code>	integer	Pixel X of the top-left corner
<code>y1</code>	integer	Pixel Y of the top-left corner
<code>x2</code>	integer	Pixel X of the bottom-right corner
<code>y2</code>	integer	Pixel Y of the bottom-right corner

Seat object

Field	Type	Description
<code>seat_id</code>	string	Stable unique identifier for the seat
<code>x</code>	integer	Pixel X of the seat's centre point, where top-left corner of the sensor's coverage area is (0, 0).
<code>y</code>	integer	Pixel Y of the seat's centre point, where top-left corner of the sensor's coverage area is (0, 0).
<code>assigned</code>	boolean	<code>true</code> if this seat is assigned to an organization, <code>false</code> otherwise

<code>organization_code</code>	string null	Code for the occupying organization. <code>null</code> if assigned is <code>false</code>
<code>organization_name</code>	string null	Full name of the occupying organization. <code>null</code> if assigned is <code>false</code>

Error responses

HTTP status	<code>error</code> value	Meaning
404	<code>"sensor_not_found"</code>	No sensor with that name exists on the specified building and floor

```
{ "error": "sensor_not_found", "message": "No sensor named 'Sensor1' found on floor 3 of Al-Midra Tower." }
```